

Gridworld Example 3.8 (Sutton & Barto): Python Simulation and Solution

1 The problem

5×5 gridworld, equiprobable random policy, off-grid actions give reward −1 and leave the agent in place, on-grid actions give reward 0. Discount factor $\gamma = 0.9$. Compute $v_\pi(s)$.

2 Bellman equation

$$v_\pi(s) = \frac{1}{4} \sum_{a \in \{N, S, E, W\}} [r(s, a) + \gamma v_\pi(s')].$$

3 Iterative policy evaluation

$$v_{k+1}(s) \leftarrow \frac{1}{4} \sum_a [r(s, a) + \gamma v_k(s')].$$

4 Python code

```
1 import numpy as np
2
3 N_ROWS, N_COLS = 5, 5
4 GAMMA = 0.9
5 THETA = 1e-6
6 ACTIONS = ['N', 'S', 'E', 'W']
7 ACTION_DELTAS = {'N': (-1, 0), 'S': (+1, 0), 'E': (0, +1), 'W': (0, -1)}
8
9 class Gridworld:
10     def __init__(self, n_rows=5, n_cols=5, gamma=0.9):
11         self.n_rows, self.n_cols, self.gamma = n_rows, n_cols, gamma
12
13     def step(self, state, action):
14         dr, dc = ACTION_DELTAS[action]
15         r, c = state
16         nr, nc = r + dr, c + dc
17         if 0 <= nr < self.n_rows and 0 <= nc < self.n_cols:
18             return (nr, nc), 0.0
19         else:
20             return (r, c), -1.0
21
22     def all_states(self):
23         return [(r, c) for r in range(self.n_rows) for c in range(self.n_cols)]
24
25 def iterative_policy_evaluation(env, theta=1e-6):
26     states = env.all_states()
27     v = {s: 0.0 for s in states}
```

```

28     while True:
29         delta = 0.0
30         v_new = {}
31         for s in states:
32             total = sum(env.step(s, a)[1] + env.gamma * v[env.step(s, a)[0]]
33                        for a in ACTIONS)
34             v_new[s] = total / len(ACTIONS)
35             delta = max(delta, abs(v_new[s] - v[s]))
36         v = v_new
37         if delta < theta:
38             break
39     return v
40
41 if __name__ == "__main__":
42     env = Gridworld()
43     v = iterative_policy_evaluation(env)
44     for r in range(env.n_rows):
45         print(" | ".join(f"{v[(r, c)]:5.1f}" for c in range(env.n_cols)))

```

5 Expected output

```

3.3 | 8.8 | 4.4 | 5.3 | 1.5
1.5 | 3.0 | 2.3 | 1.9 | 0.5
0.1 | 0.7 | 0.7 | 0.4 | -0.4
-1.0 | -0.4 | -0.4 | -0.6 | -1.2
-1.9 | -1.3 | -1.2 | -1.4 | -2.0

```

6 Corner cell calculation

For state $(0, 0)$:

$$v(0, 0) = \frac{1}{4} [2(-1 + 0.9v(0, 0)) + 0.9v(1, 0) + 0.9v(0, 1)].$$

With $v(1, 0) = 1.5$ and $v(0, 1) = 8.8$:

$$4v(0, 0) = 7.27 + 1.8v(0, 0) \implies v(0, 0) = 3.3045.$$